MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATIONS, OCTOBER 2025

Third Semester

DSE-MG3DSEZGY200

Ethology

Duration: 2 hours Total Marks: 70

PART A – MCQ (12 × 1 mark) Writeall12. (12 X 1 =12 marks)

1. Which scientist is associated with the concept of imprinting in ethology? [K][CO:1]

a) Charles Darwin
b) Ivan Pavlov
c) Konrad Lorenz
d) Karl von Frisch

2. Identify part of the brain that is primarily involved in forming long-term memories [U][CO:2]
a) Hypothalamus
b) Hippocampus
c) Amygdala
d) Cerebellum

3. Which learning model was proposed by Lorenz to explain motivation in animals? [K][CO:2]

a) Classical Conditioning
b) Cognitive Map Model
c) Psycho-Hydraulic Model
d) Learning Set Model

4. Which of the following is an example of non-associative learning? [U][CO:2]

a) Classical Conditioning
b) Operant Conditioning
c) Habituation
d) Instrumental learning

5. Namethe brain area that is primarily involved in speech production [U][CO:2]

a) Wernicke's area
b) Hippocampus

c) Amygdala
d) Broca's area

6. Which of the following is a type of sexual selection involving elaborate displays to attract mates? [K][CO:3]

a) Parental investment
b) Imprinting
c) Courtship behavior
d) Foraging behavior

7. Identify the physiological adaptation for surviving winter by reducing metabolic rate? [U][CO:3]

a) Aestivation
b) Homing
c) Hibernation
d) Navigation

8. Recall the term that describes helping behavior which benefits relatives, enhancing inclusive fitness. [K][CO:4]

a) Reciprocal altruism
b) Kin selection
c) Aggregation
d) Group selection

9. Which type of communication involves the release of chemical signals between individuals? [U][CO:5]

a) Visual
b) Tactile
c) Olfactory
d) Auditory

10. What type of pheromone is responsible for trail marking in ants? [U][CO:5]

a) Sex pheromone
b) Alarm pheromone
c) Trail pheromone
d) Primer pheromone

II. PART B – Short Answer (3 marks) Writeany10. (10 X3 =30marks)

11. Describe ethogram and its significance in behavioral studies. [U][CO:1]

12. Explain the structure and function of the hippocampus in learning and memory. [U][CO:2]
13. Explain Lorenz's psycho-hydraulic model of motivation. [U][CO:2]
14. Differentiate between habituation and sensitization with examples. [U][CO:2]
15. What is the role of sleep in memory consolidation? [U][CO:2]
16. Explain parental care in stickleback fish. [U][CO:3]
17. What is biological rhythm? Mention any two types with examples. [U][CO:3]
18. Describe the physiological basis of stress. [U][CO:3]
19. Define mimicry. Differentiate between Batesian and Müllerian mimicry. [An][CO:3]
20. Explain kin selection and its significance in social animals. [U][CO:4]
21. Write a note on territoriality. Mention two examples of territorial behavior. [K][CO:4]
22. Describe the bee dance and its role in communication among honeybees. [U][CO:5]

PART C – Short Essay (5 marks) Writeany 6. (6 X5 =30marks)

23. Explain the different types of animal communication with suitable examples. How do these help in survival and reproduction? [U][CO:5]
24. Describe the neural mechanisms and brain regions involved in learning and behavior. [U][CO:2]
25. Discuss different mating systems in animals and explain sexual selection with examples. [U][CO:3]
26. Compare various types of mimicry and their adaptive advantages in animals. [U][CO:3]
27. List out the benefits and costs of group living in animals. Explain with examples. [U][CO:4]
28. Describe the social organization in ants or honeybees. What makes their behavior eusocial? [U][CO:4]
29. Explain the different types of learning in animals with suitable examples.[U][CO:2]
30. Give an account of biological rhythms and their significance in animal behavior. [U][CO:3]